# Salience and Cryptocurrency Returns: Evidence from Bitcoin-Based Predictive Signals*

**Abstract**

Salience Theory predicts that outcomes that stand out relative to a reference receive excess weight in beliefs. Motivated by the prominence of Bitcoin in investor attention, we take Bitcoin as the reference and measure salience relative to it. We construct two salience measures, a return-based index (STR) and a novel volume-based index (STV), and confirm that both show negative predictive power for subsequent returns. The STR effect is concentrated among smaller coins, whereas STV is stronger for larger and more liquid assets and remains robust to standard controls and factor-spanning tests. Salience factors do not subsume canonical crypto anomalies, indicating a distinct behavioral dimension rather than a replacement for market, size, or momentum effects.

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## 1. Introduction

Asset pricing in the cryptocurrency market poses unique challenges. As an emerging and highly volatile asset class, cryptocurrencies lack conventional fundamentals such as cash flows or earnings, making valuation heavily dependent on market-driven variables like returns, trading volume, and liquidity (Yuan & Wang, 2018). Prices are also strongly influenced by sentiment, technological developments, regulation, and global macroeconomic conditions (Makarov & Schoar, 2020). In such an environment, behavioral forces can play a first-order role.

Among behavioral frameworks, Salience Theory (ST) offers a tractable mechanism for systematic mispricing. Bordalo et al. (2012, 2013) posit that decision weights are distorted toward attention-grabbing states, so that outcomes that "stand out" relative to a reference receive excess weight in beliefs. Empirical applications in equities typically implement ST relative to market-wide benchmarks, such as the cross-sectional average return or the market portfolio (e.g., Cosemans & Frehen, 2021; Cakici & Zaremba, 2022). In crypto, aggregate market indices can indeed be constructed (e.g., Cai & Zhao, 2024; Babiak & Bianchi, 2025), but coverage, weighting, and exchange selection vary across providers, and no single index serves as a universally recognized focal benchmark. Because ST is fundamentally benchmark-relative, and because investors in practice tend to focus on concrete, highly visible references rather than an abstract average, we use Bitcoin, the dominant and most closely followed cryptocurrency whose price and volume are prominently displayed on virtually all trading interfaces, as the reference point for computing salience. Methodologically, our implementation keeps the core Bordalo et al. (2013) structure of state-dependent, salience-weighted expectations while replacing the reference payoff with Bitcoin's performance.

We extend ST along a dimension that is particularly relevant for digital assets: trading volume. Volume captures both liquidity and investor attention; it signals market participation and information flow, but it also amplifies visibility and reactions (Karpoff, 1987; Lo & Wang, 2000; King & Koutmos, 2021). In cryptocurrencies, volume fluctuations may matter even more than returns because they reflect speculative trading, transaction utility, and perceived market credibility (Cong et al., 2023; Trozze et al., 2022). Building on these insights, we measure salience relative to Bitcoin and construct two measures: a return-based measure (STR) and a novel volume-based measure (STV) that quantifies how a coin's trading activity "stands out" against this reference.

Using daily and monthly data from *CoinMarketCap* spanning January 2016 to January 2025, consolidated across 250+ spot exchanges, we apply standard quality filters and survivorship adjustments in the spirit of Babiak & Bianchi (2025).[1] We compute STR and STV each month from within-month daily states and evaluate their pricing content using three complementary tests: (i) quintile portfolio sorts, (ii) monthly Fama–MacBeth cross-sectional regressions with standard controls (size, momentum, beta, short-term reversal, MAX/MIN, idiosyncratic volatility, turnover, skewness, downside beta, and idiosyncratic skewness), and (iii) factor-spanning tests that form salience-based factors from STR and STV and run time-series regressions of high–low anomaly portfolios on MKT, SIZE, MOM with and without the salience factor(s).

Three empirical findings emerge. *First*, in univariate portfolio tests, STR displays a strong and nearly monotonic *negative* premium under equal weighting: low-salience portfolios earn about 6.3% per month while high-salience portfolios earn roughly −0.7%, producing a highly significant High–Low spread of −7.0% and a three-factor alpha spread of −5.8%. Under value weighting, STR spreads shrink and lose significance, indicating that return-based salience is most pronounced among smaller or less-weighted coins. STV, by contrast, is weak under equal weighting but becomes economically and statistically meaningful under value weighting: the High–Low raw return spread is about −4.1% per month with a corresponding alpha around −4.2%, pointing to a volume-driven salience penalty concentrated in larger, more liquid assets.[2] *Second*, monthly Fama–MacBeth regressions confirm robust *negative* slopes for both measures. STR is strongly negative in sparse specifications; STV remains negative and statistically significant when entered alongside STR and a rich set of controls, and both measures carry incremental information in saturated models. *Third*, when we form salience-based factors and test them against a broad set of crypto anomalies in time-series regressions, neither STR nor STV fully subsumes the anomaly alphas (in contrast to Liu et al. (2022) for standard factors), suggesting that salience represents an additional behavioral pricing component rather than a replacement for size, momentum, or volume-related effects.

This paper makes two contributions while delivering a unified set of findings. We adapt Salience Theory by measuring salience relative to Bitcoin, a practical reference that reflects how crypto investors process information, and we extend the framework by introducing a volume-based salience measure (STV) alongside the standard

[1] At month $t$ we exclude assets with market capitalization below USD 1 million at $t-1$ and observations with missing prices, returns, market capitalization, or trading volume. Following Babiak & Bianchi (2025), we retain delisted coins that have at least six months of transactions and remove daily observations with non-positive closing prices. We truncate daily returns below −100% or above +150%, eliminating fewer than 0.5% of extreme or erroneous observations.

[2] We replace Bitcoin with the cross-sectional average as the reference leaves STR largely unchanged but makes STV weak and often non-monotonic. This implies that volume-based salience contains pricing information primarily when measured relative to Bitcoin.

return-based measure (STR). Empirically, both measures predict lower subsequent returns in portfolio tests and in Fama–MacBeth regressions with rich controls; the effects are incremental to one another and robust to factor-spanning tests. Salience factors do not subsume canonical crypto anomalies, indicating a distinct behavioral dimension rather than a replacement for market, size, or momentum effects. Overall, the evidence is consistent with salience-driven mispricing in cryptocurrencies: when salience is measured relative to Bitcoin, salient states receive disproportionate decision weight, generating systematic overvaluation of high-salience assets and lower subsequent expected returns.

## 2. Data and Methodology

### *2.1. Data source*

Following Liu et al. (2022) and Bianchi et al. (2022), we assemble a panel of crypto assets from *CoinMarketCap*, using daily and monthly OHLCV and market capitalization consolidated across more than 250 spot exchanges. To ensure sample quality and mitigate liquidity-driven return errors and survivorship bias, we apply several filters. At month $t$, we exclude assets with market capitalization below USD 1 million at $t-1$ and observations with missing prices, returns, market capitalization, or trading volume. Following Babiak & Bianchi (2025), we retain delisted coins provided they exhibit at least six months of transaction history, remove daily observations with non-positive closing prices, and truncate extreme daily returns below $-100\%$ or above $+150\%$, which eliminates fewer than 0.5% of erroneous or extreme observations. Daily data are used to construct within-month state variables, while portfolio formation and cross-sectional regressions are conducted at the monthly frequency. Unless otherwise noted, all characteristics are winsorized at the 1st and 99th percentiles each month. After applying these filters, the pre-2016 cross-section is too small for reliable portfolio and Fama–MacBeth tests (for example, there are only about 40–47 assets in 2014–2015, compared with 80+ in 2016 and 288+ in 2017). Accordingly, our baseline sample spans January 2016 to January 2025.[3]

### *2.2. Salience-based measures*

We operationalize salience following Bordalo et al. (2013) and Cosemans & Frehen (2021). In their framework, each state $s$ is characterized by an asset payoff and a reference payoff, and the state is more *salient* the more strongly the assets payoff stands out relative to the reference level. Formally, salience is increasing in the absolute difference between the two payoffs, scaled by their magnitudes, and more salient states receive overweighted decision weights. In equity applications, the reference payoff is typically taken to be the market portfolio or the cross-sectional average return, implicitly assuming a well-defined "average" benchmark.

In cryptocurrency markets, aggregate "market" returns can be and have been computed, such as value-weighted indices in Babiak & Bianchi (2025) and related studies. However, Salience Theory is fundamentally benchmark-relative: perceptions are formed against a focal, attention-grabbing reference rather than an abstract cross-sectional average. Unlike equities, where the market portfolio is a widely accepted benchmark, crypto lacks a single, universally recognized reference index (coverage, weights, and exchange selection vary across providers). In this

[3] This window covers major episodes in digital-asset markets, including the late-2017 ICO boom, the 2018–2019 "crypto winter," the March 2020 COVID-19 shock, and the 2021–2022 boom and subsequent retracement, as well as institutional changes such as the introduction of Bitcoin and Ether futures and Ethereums transition from proof-of-work to proof-of-stake.

environment, investors are more likely to focus on a concrete, highly visible asset. Because Bitcoin is the dominant and most closely followed cryptocurrency, with price and volume prominently displayed on virtually all trading interfaces, we measure salience relative to Bitcoins performance. Our implementation follows Bordalo et al. (2012, 2013) in using state-dependent, salience-weighted decision weights; relative to equity implementations that reference the market average (for example, Cosemans & Frehen, 2021), we retain the same functional form and weighting scheme while taking Bitcoins performance as the reference payoff.

For asset $i$ in month $t$, let $S_t$ denote the set of trading days and index states by $s \in S_t$. Let $r_{i,s,t}$ be the assets daily return and $r^{\mathrm{BTC}}_{s,t}$ Bitcoins daily return on day $s$ in month $t$. We define the salience of the assets daily return relative to Bitcoin as

$$\sigma\left(r_{i,s,t}, r^{\mathrm{BTC}}_{s,t}\right) = \frac{\left|r_{i,s,t} - r^{\mathrm{BTC}}_{s,t}\right|}{|r_{i,s,t}| + \left|r^{\mathrm{BTC}}_{s,t}\right| + \theta}, \qquad \theta = 0.1. \tag{1}$$

Rank states by $\sigma$ from most to least salient and denote the rank by $k_{is,t} \in \{1, \ldots, |S_t|\}$. With objective (uniform) probabilities $\pi_{s,t} = |S_t|^{-1}$, the distorted subjective weights are

$$\tilde{\pi}_{is,t} = \frac{\pi_{s,t}\, \delta^{k_{is,t}}}{\sum_{s' \in S_t} \pi_{s',t}\, \delta^{k_{is',t}}}, \qquad \delta = 0.7. \tag{2}$$

Our return-based salience measure for asset $i$ in month $t$ is the covariance across states between the weight distortion and realized returns,

$$\mathrm{STR}_{i,t} = \mathrm{cov}_{s \in S_t}\left(\frac{\tilde{\pi}_{is,t}}{\pi_{s,t}}, r_{i,s,t}\right), \tag{3}$$

which we interpret as *salience-through-returns* relative to Bitcoin.

Analogously, we construct a volume-based salience measure (*salience-through-volume*) using trading activity. Let $v_{i,s,t}$ be the assets trading volume and $v^{\mathrm{BTC}}_{s,t}$ Bitcoins trading volume on day $s$. Define

$$\sigma_v\left(v_{i,s,t}, v^{\mathrm{BTC}}_{s,t}\right) = \frac{\left|v_{i,s,t} - v^{\mathrm{BTC}}_{s,t}\right|}{|v_{i,s,t}| + \left|v^{\mathrm{BTC}}_{s,t}\right| + \theta}, \tag{4}$$

obtain the corresponding ranks $k^v_{is,t}$ and distorted weights

$$\tilde{\pi}^v_{is,t} = \frac{\pi_{s,t}\, \delta^{k^v_{is,t}}}{\sum_{s' \in S_t} \pi_{s',t}\, \delta^{k^v_{is',t}}}, \tag{5}$$

and define

$$\mathrm{STV}_{i,t} = \mathrm{cov}_{s \in S_t}\left(\frac{\tilde{\pi}^v_{is,t}}{\pi_{s,t}}, v_{i,s,t}\right). \tag{6}$$

Both STR and STV are computed at the monthly frequency and enter our portfolio sorts and Fama–MacBeth regressions.

### 2.3. *Control variables*

At the monthly frequency, we include a set of standard crypto/equity-style characteristics. Size is defined as log(ME), where ME denotes the end-of-month market capitalization. Short-term reversal $\mathrm{REV}_{i,t-1}$ is the lagged one-month return, while momentum $\mathrm{MOM}_{i,t-1}$ is the cumulative return over the past 3 months excluding the most recent month. Market beta (BETA) and idiosyncratic volatility (IVOL) are obtained from a market-model regression of monthly returns, and downside beta (DBETA) follows Ang et al. (2006) using days with market returns below their 12-month mean. We also include MAX and MIN (the monthly maximum and minimum daily returns; Bali et al., 2011), total skewness (SKEW) and idiosyncratic skewness (ISKEW; Boyer et al., 2010), and turnover (TK), measured as total monthly dollar trading volume divided by end-of-month market capitalization. All controls are updated at the monthly frequency and winsorized at the 1st and 99th percentiles.

## 3. Results of empirical tests

### *3.1. Univariate portfolio analysis*

We study the pricing role of salience in crypto assets using portfolio sorts on the return-based measure (STR) and the volume-based measure (STV). Table 1 shows a strong and monotonic pattern for STR: under equal weighting, average returns decline from about 6.3% in the low-salience portfolio to $-0.7\%$ in the high-salience portfolio, yielding a High–Low spread of $-7.0\%$ per month (High–Low $= -0.070$, $t = -7.165$) and a three-factor alpha spread of $-5.8\%$ (High–Low $= -0.058$, $t = -8.574$), both highly significant. Under value weighting, average returns cluster between roughly 7.8% and 11.6%, and the High–Low spreads in raw returns and alphas (about $-2.0\%$ and $-1.9\%$ per month, with $t$-statistics around $-1.4$ to $-1.6$) are no longer statistically significant, indicating that return-based salience primarily predicts lower future returns among smaller or less-weighted coins. For STV, equal-weighted patterns are weak, whereas value-weighted portfolios display economically and statistically meaningful penalties for the most salient coins. As an alternative reference-point specification, we also recompute salience using the cross-sectional average market return (or average trading volume) as the reference benchmark: STR delivers High–Low spreads similar in magnitude and significance to those in Table 1, while the volume-based specification using average market volume is much weaker and often non-monotonic, with small and statistically insignificant alpha spreads.[4]

For the volume-based measure STV, the evidence is more nuanced. Equal-weighted raw returns drift only mildly upward from about 4.1% to 4.7% across the STV quintiles, and the EW high–low spread in raw returns is small and insignificant (High–Low $= 0.006$, $t = 0.506$). EW alphas are negative and sometimes significant for the low- and mid-STV portfolios (around $-2.9\%$ to $-4.2\%$ per month), but do not display a clear monotonic pattern, and the high–low alpha spread of 2.1% per month (High–Low $= 0.021$, $t = 1.633$) remains statistically insignificant. In value-weighted terms, however, there is a pronounced underperformance of the most salient coins: VW raw returns rise from 12.6% for low-STV assets to about 14% for intermediate portfolios, but fall to 8.5% for the high-STV portfolio, yielding a VW high–low raw return spread of $-4.1\%$ (High–Low $= -0.041$, $t = -2.179$). VW alphas exhibit a similar pattern, decreasing from roughly 4.9% to 0.7% per month, with a high–low alpha spread of $-4.2\%$ (High–Low $= -0.042$, $t = -1.896$). Overall, the quintile portfolio evidence points to a robust negative pricing of return-based salience and weaker, but still suggestive, penalties for extreme volume-based salience once large, liquid coins are given more weight.

[4]Due to space limitations, we do not present the results in the main text; however, these results can be provided upon request.

Table 1: Quintile portfolios sorted on salience measures

| | Return-based salience (STR) | | | | Volume-based salience (STV) | | | |
|---|---|---|---|---|---|---|---|---|
| Portfolio | EW $R$ | EW $\alpha$ | VW $R$ | VW $\alpha$ | EW $R$ | EW $\alpha$ | VW $R$ | VW $\alpha$ |
| Low ST | 0.063* | -0.004 | 0.098*** | 0.019 | 0.041 | -0.029* | 0.126*** | 0.049** |
| | (1.858) | (-0.310) | (2.670) | (1.451) | (1.266) | (-1.807) | (3.095) | (2.306) |
| 2 | 0.052 | -0.011 | 0.096*** | 0.039** | 0.025 | -0.042*** | 0.107*** | 0.038*** |
| | (1.613) | (-0.876) | (3.080) | (2.265) | (0.801) | (-3.036) | (3.177) | (2.912) |
| 3 | 0.050 | -0.016 | 0.089** | 0.014 | 0.037 | -0.029* | 0.141*** | 0.077*** |
| | (1.508) | (-1.182) | (2.554) | (0.957) | (1.063) | (-1.886) | (3.218) | (3.369) |
| 4 | 0.037 | -0.027** | 0.116*** | 0.033** | 0.044 | -0.011 | 0.143*** | 0.085*** |
| | (1.272) | (-2.182) | (3.296) | (2.056) | (1.430) | (-0.702) | (3.894) | (3.901) |
| High ST | -0.007 | -0.063*** | 0.078** | 0.001 | 0.047* | -0.008 | 0.085** | 0.007 |
| | (-0.248) | (-4.300) | (2.177) | (0.042) | (1.718) | (-0.872) | (2.530) | (1.592) |
| High–Low | -0.070*** | -0.058*** | -0.020 | -0.019 | 0.006 | 0.021 | -0.041** | -0.042* |
| | (-7.165) | (-8.574) | (-1.400) | (-1.591) | (0.506) | (1.633) | (-2.179) | (-1.896) |

*Notes:* This table reports next-month returns on quintile portfolios sorted on the return-based salience measure (STR) and the volume-based salience measure (STV). At the end of each month, all cryptocurrencies are sorted into five portfolios by STR (or STV) and held for month $t+1$. "EW" and "VW" denote equal-weighted and value-weighted portfolios. $R$ denotes the average next-month *raw* portfolio return; $\alpha$ denotes the monthly three-factor alpha relative to the crypto market, size, and momentum factors. Newey–West $t$-statistics (in parentheses) are reported for both $R$ and $\alpha$. High–Low is the return on the highest-salience portfolio minus the lowest-salience portfolio. Returns and alphas are reported in decimal form. $^{*}$, $^{**}$, and $^{***}$ indicate significance at the 10%, 5%, and 1% levels, respectively.

### 3.2. *Fama–MacBeth regression analysis*

To isolate the partial effect of salience, we estimate monthly Fama–MacBeth regressions (Fama & MacBeth, 1973) of the form

$$r_{i,t+1} = \lambda_{0,t} + \lambda_{1,t}\,\mathrm{ST}_{i,t} + \lambda_{2,t}^{\top} W_{i,t} + v_{i,t}, \tag{7}$$

where $\mathrm{ST}_{i,t} \in \{\mathrm{STR}_{i,t}, \mathrm{STV}_{i,t}\}$ and $W_{i,t}$ collects standard characteristics (size, momentum, market beta, short-term reversal, MAX/MIN, idiosyncratic volatility, turnover, skewness, downside beta, and idiosyncratic skewness). When STR is the regressor of interest, we include STV in $W_{i,t}$, and vice versa. All right-hand-side variables are standardized each month to have zero mean and unit variance, so coefficients can be interpreted as the effect of a one-standard-deviation change in the corresponding characteristic.

Table 2 reports the average slope coefficients and their Newey–West $t$-statistics. Columns (1)–(3) show that the return-based salience measure STR loads negatively and significantly in univariate specifications. The coefficient is about $-0.011$ in column (1) (High–Low effect of roughly $-1.1\%$ per month, $t = -4.100$) and remains statistically significant at the 1% level when adding size and momentum controls. Including short-term reversal in column (3) reduces the magnitude to around $-0.007$ (still significant at the 1% level, $t = -2.587$), indicating that part of the salience effect is related to short-horizon return dynamics but is not fully explained by them.

Columns (4)–(6) document a similar pattern for the volume-based measure STV. The coefficient is about $-0.004$ across these specifications, significant at the 5% level, and largely unchanged by the inclusion of momentum and reversal. Thus, both return- and volume-based salience predict lower next-month returns in simple and moderately controlled specifications, with the STR premium somewhat larger in magnitude.

When we allow the two salience measures to compete in the same regression, their relative roles become clearer. In column (7), where both STR and STV enter alongside size, momentum, and reversal, STV remains strongly significant (coefficient $-0.003$, $t = -2.815$), whereas the STR coefficient shrinks to $-0.002$ and becomes insignificant. Adding the full set of controls in columns (8)–(10) shows that the volume-based effect is particularly robust to the inclusion of additional characteristics: in column (9), where only STV and the full control set enter, the coefficient on STV is $-0.003$ ($t = -2.694$), significant at the 1% level, while STR is no longer significant when included alone with the same controls (column (8)). In the most saturated specification (column (10)), which includes both salience measures and all controls, the coefficients on STR and STV are $-0.009$ ($t = -3.238$) and $-0.004$ ($t = -1.925$), respectively, indicating that return- and volume-based salience each carry incremental information about expected returns.

Table 2: Fama–MacBeth regressions of next-month returns on salience measures

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) |
|---|---|---|---|---|---|---|---|---|---|---|
| STR | $-0.011^{***}$ | $-0.009^{***}$ | $-0.007^{***}$ | | | | $-0.002$ | $-0.002$ | | $-0.009^{***}$ |
| | $(-4.100)$ | $(-3.447)$ | $(-2.587)$ | | | | $(-0.801)$ | $(-0.786)$ | | $(-3.238)$ |
| STV | | | | $-0.004^{**}$ | $-0.004^{**}$ | $-0.004^{**}$ | $-0.003^{***}$ | | $-0.003^{***}$ | $-0.004^{*}$ |
| | | | | $(-2.283)$ | $(-2.042)$ | $(-2.254)$ | $(-2.815)$ | | $(-2.694)$ | $(-1.925)$ |
| ME | $-0.005$ | $-0.004$ | $-0.004$ | $-0.007$ | $-0.005$ | $-0.005$ | $-0.008^{**}$ | $-0.007^{*}$ | $-0.007^{**}$ | $-0.005$ |
| | $(-1.094)$ | $(-1.185)$ | $(-1.274)$ | $(-1.440)$ | $(-1.311)$ | $(-1.358)$ | $(-2.179)$ | $(-1.916)$ | $(-1.985)$ | $(-1.329)$ |
| MOM | | $0.006^{*}$ | $0.004$ | | $0.007^{**}$ | $0.004$ | $0.003$ | $0.003$ | $0.003$ | $0.006^{*}$ |
| | | $(1.737)$ | $(1.406)$ | | $(2.155)$ | $(1.477)$ | $(0.896)$ | $(0.886)$ | $(0.821)$ | $(1.712)$ |
| REV | | | $-0.007$ | | | $-0.009^{**}$ | $-0.005$ | $-0.005$ | $-0.006^{*}$ | |
| | | | $(-1.390)$ | | | $(-2.243)$ | $(-1.310)$ | $(-1.329)$ | $(-1.790)$ | |
| BETA | | | | | | | | $0.002$ | $0.002$ | $0.002$ |
| | | | | | | | | $(0.549)$ | $(0.589)$ | $(0.676)$ |
| MAX | | | | | | | | $-0.006$ | $-0.007$ | $-0.008$ |
| | | | | | | | | $(-1.010)$ | $(-1.104)$ | $(-1.416)$ |
| MIN | | | | | | | | $0.005$ | $0.005$ | $0.004$ |
| | | | | | | | | $(0.735)$ | $(0.674)$ | $(0.575)$ |
| IVOL | | | | | | | | $-0.007^{**}$ | $-0.007^{**}$ | $-0.007^{**}$ |
| | | | | | | | | $(-2.381)$ | $(-2.435)$ | $(-2.506)$ |
| TK | | | | | | | | $-0.003$ | $-0.002$ | $-0.003$ |
| | | | | | | | | $(-0.552)$ | $(-0.521)$ | $(-0.709)$ |
| SKEW | | | | | | | | $-0.000$ | $0.001$ | $-0.000$ |
| | | | | | | | | $(-0.058)$ | $(0.143)$ | $(-0.061)$ |
| DBETA | | | | | | | | $0.008$ | $0.007$ | $0.007$ |
| | | | | | | | | $(0.986)$ | $(0.899)$ | $(0.869)$ |
| ISKEW | | | | | | | | $0.003$ | $0.003$ | $0.004$ |
| | | | | | | | | $(1.222)$ | $(1.177)$ | $(1.315)$ |

*Notes:* This table reports Fama–MacBeth cross-sectional regressions of next-month cryptocurrency returns on the return-based salience measure (STR), the volume-based salience measure (STV), and other characteristics. All right-hand-side variables are standardized to have zero mean and unit variance each month. Columns (1)–(3) include STR with progressively richer controls; columns (4)–(6) repeat the exercise for STV; column (7) includes both STR and STV with a basic control set; columns (8) and (9) add the full set of controls separately for STR and STV; column (10) includes both STR and STV with all controls. Coefficients are monthly and can be interpreted as the effect of a one-standard-deviation increase in the corresponding characteristic on expected next-month returns. Newey–West $t$-statistics (using 4 lags) are reported in parentheses. $^{*}$, $^{**}$, and $^{***}$ indicate significance at the 10%, 5%, and 1% levels, respectively.

Among the control variables, size (ME) generally loads negatively, consistent with a size premium, and is statistically significant in several specifications once additional controls are added. Momentum (MOM) shows a positive and sometimes significant effect, while short-term reversal (REV) tends to have a negative coefficient and becomes significant in some specifications, consistent with partial return continuation at intermediate horizons and mild reversal at very short horizons. Idiosyncratic volatility (IVOL) is negatively related to subsequent returns and remains significant after controlling for salience, whereas other higher-moment and trading-activity measures (MAX, MIN, TK, SKEW, DBETA, ISKEW) are generally insignificant. Overall, the Fama–MacBeth evidence points to a robust, economically meaningful salience premium in cryptocurrencies, with volume-based salience especially resilient to the inclusion of a rich set of controls.

### 3.3. *Salience factors and crypto return anomalies*

If salience captures a key trading motive in crypto markets, a salience-based factor should help organize other return "anomalies." Following Liu et al. (2022), we construct factor-mimicking portfolios for return-based and volume-based salience and test whether these factors help explain characteristic-sorted hedge returns. At each rebalancing date, we sort all cryptocurrencies into three salience groups using either STR or STV (bottom 30%, middle 40%, top 30%), form value-weighted portfolios, and define the salience factor as the up-minus-down (H–L) return. By construction, a more negative H–L corresponds to a stronger salience premium.

We evaluate whether these salience factors account for a broad set of crypto anomalies by running time-series regressions of H–L excess returns under four benchmark specifications,

$$\text{(M1)} \quad R_i - R_f = \alpha_i + \beta_{i,1}\,\text{MKT} + \beta_{i,2}\,\text{SIZE} + \beta_{i,3}\,\text{MOM} + \varepsilon_i, \tag{8}$$

$$\text{(M2)} \quad R_i - R_f = \alpha_i + \gamma_{i,1}\,\text{ST}^X + \varepsilon_i, \tag{9}$$

$$\text{(M3)} \quad R_i - R_f = \alpha_i + \beta_{i,1}\,\text{MKT} + \beta_{i,2}\,\text{SIZE} + \gamma_{i,1}\,\text{ST}^X + \varepsilon_i, \tag{10}$$

$$\text{(M4)} \quad R_i - R_f = \alpha_i + \beta_{i,1}\,\text{MKT} + \beta_{i,2}\,\text{SIZE} + \beta_{i,3}\,\text{MOM} + \gamma_{i,1}\,\text{ST}^X + \varepsilon_i, \tag{11}$$

where $\text{ST}^X$ denotes either the STR-based or the STV-based salience factor.

Table 3 reports intercepts for both salience-factor choices estimated on the same test assets. With STR as the pricing factor, the benchmark model with MKT, SIZE, and MOM (M1) leaves most anomaly alphas economically moderate and often statistically weak, and the STV sorted $H - L$ portfolio is essentially priced. Adding STR alone (M2) does not eliminate the cross section: PRC and MAXDPRC retain negative alphas, and the volume strategies (PRCVOLM, STDPRCVOL) remain economically negative with borderline $t$-statistics. When STR augments MKT and SIZE (M3) and then the full benchmark (M4), magnitudes move only partly toward zero. MOM3 is already small and becomes negligible, whereas the volume strategies continue to display negative intercepts; MCAP behaves in line with its benchmark exposures.

Results with STV as the pricing factor are similar in structure. The STR sorted $H - L$ portfolio has alphas close to zero across specifications, PRC and MAXDPRC remain negative under M2, and volume-defined strategies stay economically negative with only marginal significance. Enriching the benchmark with STV (M3) and then adding MOM (M4) reduces the already small alpha for MOM3 to approximately zero, while long horizon momentum (MOM4) and the volume strategies retain negative and borderline significant intercepts. Overall, salience implemented through either returns (STR) or trading activity (STV) adds a largely orthogonal pricing dimension that complements rather than replaces market, size, and momentum; each salience factor weakly prices the others $H - L$ portfolio, yet neither subsumes the broader suite of crypto anomalies.

## 4. Conclusion

We study how benchmark relative attention shapes cryptocurrency returns by adapting Salience Theory to measure salience relative to Bitcoin and by constructing two state-dependent measures, a return-based STR and a volume-based STV. Using daily states aggregated to monthly characteristics, both measures forecast lower subsequent returns. Portfolio tests and Fama–MacBeth regressions with rich controls confirm negative slopes, and each measure contributes incremental information about expected returns. When formed into factors and evaluated

Table 3: Anomaly alphas with STR- and STV-based salience factors

| | STR factor | | | | STV factor | | | |
|---|---|---|---|---|---|---|---|---|
| Anomaly | (1) M1 | (2) M2 | (3) M3 | (4) M4 | (5) M1 | (6) M2 | (7) M3 | (8) M4 |
| STR (H–L) | | | | | 0.009 | 0.040 | 0.011 | 0.009 |
| | | | | | (0.387) | (1.198) | (0.492) | (0.405) |
| STV (H–L) | -0.000 | -0.027 | -0.013 | -0.000 | | | | |
| | (-0.010) | (-1.072) | (-0.564) | (-0.000) | | | | |
| MCAP | -0.000 | -0.795 | 0.000*** | -0.000** | -0.000 | -0.779 | -0.000*** | 0.000 |
| | (-1.243) | (-1.528) | (3.829) | (-2.141) | (-1.296) | (-1.510) | (-2.832) | (1.168) |
| PRC | -0.038 | -0.185** | -0.063 | -0.039 | -0.038 | -0.160** | -0.051 | -0.038 |
| | (-0.912) | (-2.236) | (-1.446) | (-0.923) | (-0.972) | (-2.303) | (-1.274) | (-1.045) |
| MAXDPRC | -0.038 | -0.175** | -0.060 | -0.039 | -0.038 | -0.152** | -0.049 | -0.038 |
| | (-0.956) | (-2.198) | (-1.461) | (-0.969) | (-1.049) | (-2.258) | (-1.343) | (-1.134) |
| MOM1 | 0.014 | -0.022 | -0.006 | 0.014 | 0.014 | -0.001 | -0.002 | 0.014 |
| | (1.006) | (-0.733) | (-0.286) | (1.038) | (1.132) | (-0.042) | (-0.123) | (1.214) |
| MOM2 | 0.003 | -0.032 | -0.014 | 0.004 | 0.003 | -0.017 | -0.015 | 0.003 |
| | (0.177) | (-0.863) | (-0.553) | (0.218) | (0.187) | (-0.454) | (-0.728) | (0.188) |
| MOM3 | -0.000*** | -0.051 | -0.032 | -0.000** | -0.000*** | -0.028 | -0.027 | 0.000 |
| | (-3.373) | (-1.306) | (-1.108) | (-2.035) | (-3.058) | (-0.893) | (-1.190) | (0.000) |
| MOM4 | -0.023 | -0.073* | -0.055* | -0.024 | -0.023 | -0.049* | -0.047* | -0.023* |
| | (-1.605) | (-1.873) | (-1.943) | (-1.539) | (-1.631) | (-1.696) | (-1.929) | (-1.685) |
| PRCVOLM | -0.178 | -0.210** | -0.224* | -0.190* | -0.178 | -0.209 | -0.180 | -0.178 |
| | (-1.331) | (-2.006) | (-1.851) | (-1.731) | (-1.347) | (-1.520) | (-1.384) | (-1.348) |
| STDPRCVOL | -0.170 | -0.270** | -0.227* | -0.181 | -0.170 | -0.243* | -0.175 | -0.170 |
| | (-1.201) | (-2.001) | (-1.692) | (-1.479) | (-1.206) | (-1.682) | (-1.267) | (-1.211) |

*Notes:* Entries are estimated alphas from time-series regressions of high–low (H–L) anomaly portfolio excess returns. Columns (1)–(4) use the STR-based salience factor in M2–M4; columns (5)–(8) repeat M1–M4 using the STV-based salience factor. M1 includes MKT, SIZE, MOM; M2 includes only the salience factor; M3 adds MKT and SIZE to the salience factor; M4 adds MKT, SIZE, MOM to the salience factor. Newey–West $t$-statistics (4 lags) are in parentheses. $^{*}$, $^{**}$, and $^{***}$ denote significance at the 10%, 5%, and 1% levels.

against a broad set of crypto anomalies, neither the STR- nor the STV-based factor subsumes existing alphas, indicating that salience captures an additional behavioral dimension that complements rather than replaces market, size, and momentum effects. In some anomaly tests, the STV factor appears to organize returns somewhat better, but we do not find dominance over STR. Overall, Bitcoin serves as a practical reference for investor attention, and salience measured through either returns or trading activity helps explain the cross section of cryptocurrency returns while remaining distinct from existing factors.

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